

Hypothesis Testing I

Unit - 4
Session - 7

Unit 4: Hypothesis Testing

- Test of Mean - One sample Z Test - Test of Mean
- One sample t-Test - Test of Mean
- 2 samples - Independent and Dependent - 2 samples
- Test of proportion - One sample (Z proportion test)
- Two samples (Z proportion test)
- Case Study - Categorical vs Continuous
- More than 2 samples

Agenda

- Test of Mean
 - One sample (Z & T)
 - Independent and Dependent - 2 sample
 - Three or more samples (Anova) (details later)
- Test of proportion
 - One sample (Z proportion test)
 - 2 sample (Z proportion test)
 - Three or more samples (Chi Square) (details later)

One Sample – Z test

One-Sample test - Z test for Mean (σ known)

Z test for Mean (σ known)

$$Z_{\text{STAT}} = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

- where $\bar{X}$ is the mean of the sample
 μ is the mean of the population
 σ is the standard deviation of the population
 n is the number of observations

Decision rule

	H_1	Based on critical region	Based on p-value	Based on confidence interval
For two tailed test	$\mu \neq \mu_0$	Reject H_0 if $ Z > Z_{\alpha/2}$	Reject H_0 if p-value is less than or equal to the level of significance	Reject H_0 if μ_0 does not lie in the confidence interval
For left tailed test	$\mu < \mu_0$	Reject H_0 if $Z < -Z_\alpha$		
For right tailed test	$\mu > \mu_0$	Reject H_0 if $Z > Z_\alpha$		



Two tailed test

Question:

A car manufacturing company claims that the mileage of their new car is 25 kmpl with a standard deviation of 2.5 kmpl. A random sample of 45 cars was drawn and recorded their mileage as per the standard procedure. From the sample, the mean mileage was seen to be 24 kmpl. Is this evidence to claim that the mean mileage is different from 25 kmpl? (assume normality of data)

Test the claim using critical region technique, p-value technique and confidence interval technique. [Use $\alpha = 0.01$].



Two tailed test

Solution:

A car manufacturing company claims that the mileage of a car is 25 kmph with a standard deviation of 2.5 kmph.

Thus $\mu = 25$ kmph and $\sigma = 2.5$ kmph

A random sample of 45 cars was drawn and recorded their mileage as per the standard procedure. From the sample, the mean mileage was seen to be 24 kmph. $\bar{X}$

Thus $\bar{X} = 24$ kmph and $n = 45$

To find: Evidence to claim that the mean mileage is different from 25kmph

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Two tailed test

Solution:

Here X: the mileage of the new car

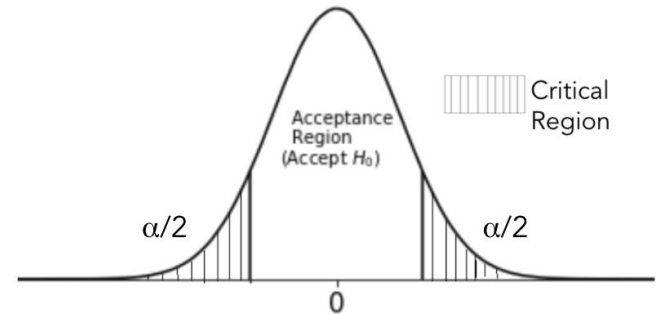
$$X \sim N(\mu, \sigma^2)$$

To test, $H_0 : \mu = 25$ against $H_1 : \mu \neq 25$

Obtain the solution using critical region technique.

The test statistics:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$



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Two tailed test

Solution:

The test statistics:

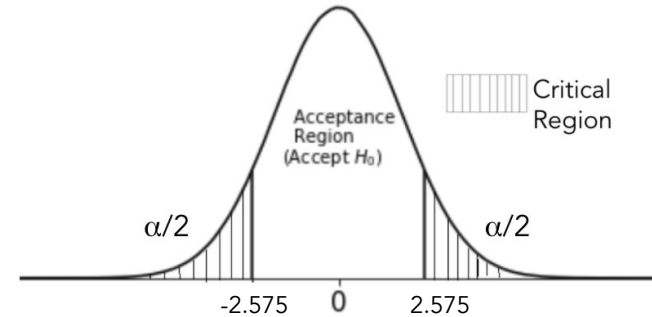
$$Z_{calc} = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} = \frac{24 - 25}{2.5 / \sqrt{45}} = -2.68$$

Here $\alpha = 0.01$, so $z_{0.005} = 2.575$

Decision: Reject H_0 , if $|Z_{calc}| > Z_{\alpha/2}$. Here $|-2.68| > 2.575$.

Thus reject H_0 .

Thus there is enough evidence to conclude that the mean mileage is different from 25 kmph.





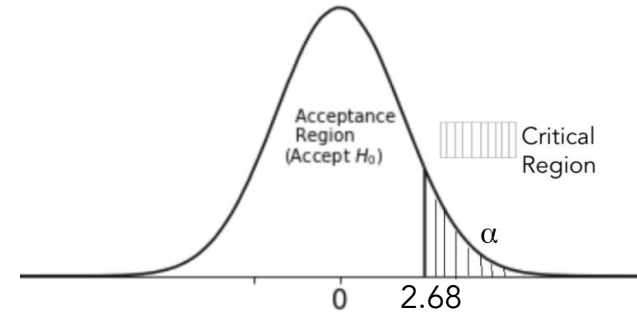
Two tailed test

Solution:

Let us obtain the decision using the p-value method.

$$Z_{calc} = -2.68$$

$$\begin{aligned}\text{The p-value} &= 2 \cdot P(|Z| > |Z_{calc}|, \text{ under } H_0) \\ &= 2 \cdot P(|Z| > 2.68, \mu = 25) \quad \dots \text{(due to symmetry)} \\ &= 0.0074\end{aligned}$$



Since p-value < 0.01 , we reject H_0 .

Thus there is enough evidence to conclude that the mean mileage is different from 25 kmph.

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Two tailed test

Solution:

Solution using confidence interval method.

The confidence interval is given by $\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

$$\begin{aligned}\text{The CI} &= \left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right) = \left(24 - 2.575 \frac{2.5}{\sqrt{45}}, 24 + 2.575 \frac{2.5}{\sqrt{45}} \right) \\ &= (23.05, 24.95)\end{aligned}$$

Since $\mu_0 = 25$, does not lie in $(23.05, 24.95)$, we reject H_0 .

Thus there is enough evidence to conclude that the mean mileage is different from 25 kmph.

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Two tailed test

Python solution: Calculate Z_{calc}

```
# define a function to calculate the Z-test statistic
# pass the population mean, population standard deviation, sample size and sample mean
def z_test(pop_mean, pop_std, n, samp_mean):

    # calculate the test statistic
    z_score = (samp_mean - pop_mean) / (pop_std / np.sqrt(n))

    # return the z-test value
    return z_score

# calculate the test statistic using the function 'z_test'
z_score = z_test(pop_mean = 25, pop_std = 2.5, n = 45, samp_mean = 24)
print("Z-score:", z_score)

Z-score: -2.6832815729997477
```

Reject H_0 , as $|Z_{\text{calc}}| > Z_{\alpha/2}$. Here $|-2.68| > 2.575$

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Two tailed test

Python solution: Calculate p-value

```
# calculate the corresponding p-value for the test statistic
# use 'cdf()' to calculate  $P(Z \leq z\_score)$ 
p_value = stats.norm.cdf(z_score)

# for a two-tailed test multiply the p-value by 2
req_p = p_value*2
print('p-value:', req_p)

p-value: 0.007290358091535638
```

Since $p\text{-value} < 0.01$, we reject H_0 .



Two tailed test

Python solution: Calculate 99% confidence interval

```
# calculate the 99% confidence interval for the population mean
# pass the sample mean to the parameter, 'loc'
# pass the scaling factor (pop_std / n^(1/2)) to the parameter, 'scale'
print('Confidence interval:', stats.norm.interval(0.99, loc = samp_mean, scale = pop_std / np.sqrt(n)))
```

Confidence interval: (23.040045096471452, 24.959954903528548)

Since $\mu_0 = 25$, does not lie in the confidence interval, we reject H_0 .



Left tailed test

Question:

A sample of 900 PVC pipes is found to have an average thickness of 12.5 mm. Can we assume that the sample is coming from a normal population with mean 13mm against that it is less than 13 mm. The population standard deviation is 1 mm. Test the hypothesis using the p-value method at 5% level of significance.



Left tailed test

Solution:

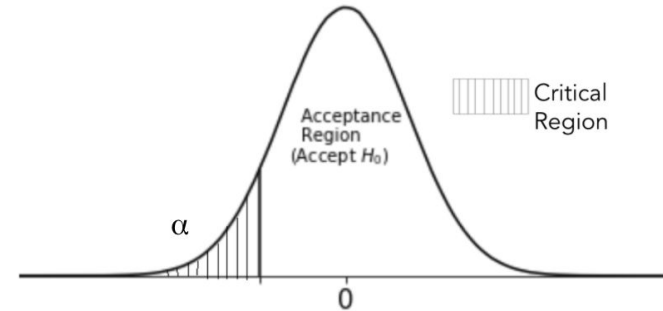
Here X: the thickness of PVC pipes

$$X \sim N(\mu, \sigma^2)$$

Here, $\mu = 13\text{mm}$, $\sigma = 1\text{ mm}$, $\bar{X} = 12.5$, $n = 900$

To test, $H_0 : \mu \geq 13$ against $H_1 : \mu < 13$

Obtain the solution using p-value technique.





Left tailed test

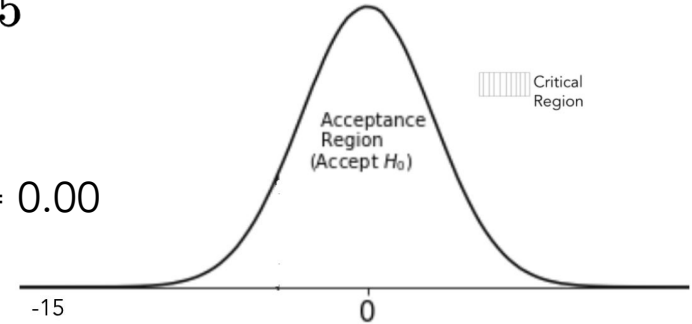
Solution:

The test statistics:
$$Z_{calc} = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} = \frac{12.5 - 13}{1 / \sqrt{900}} = -15$$

The p-value = $P(Z < Z_{calc}, \text{ under } H_0) = P(Z < -15, \mu = 13) = 0.00$

Since p-value < 0.05, we reject H_0 .

Thus there is enough evidence to conclude that the sample may not be regarded as coming from population with mean 13.





Left tailed test

Python solution: Calculate p-value

```
# calculate the corresponding p-value for the test statistic
# use 'cdf()' to calculate  $P(Z \leq -15)$ 
p_value = stats.norm.cdf(-15)

print('p-value:', p_value)

p-value: 3.6709661993126986e-51
```

Since $p\text{-value} < 0.05$, we reject H_0 .



Right tailed test

Question:

An e-commerce company claims that the mean delivery time of food items on their website in NYC is 60 minutes with a standard deviation of 30 minutes. A random sample of 45 customers ordered from the website, and the mean time for delivery was found to be 75 minutes. Is this enough evidence to claim that the mean time to get items delivered is more than 60 minutes. (assume normality of data)

Test the claim using p-value technique. [Use $\alpha = 0.05$].



Right tailed test

Solution:

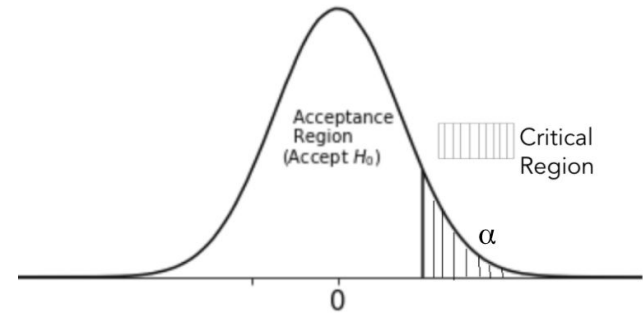
Here X: Time (in minutes) to receive the delivery

$$X \sim N(\mu, \sigma^2)$$

Here, $\mu = 60$, $\sigma = 30$, $\bar{X} = 75$, $n = 45$

To test, $H_0 : \mu \leq 60$ against $H_1 : \mu > 60$

Obtain the solution using p-value technique.





Right tailed test

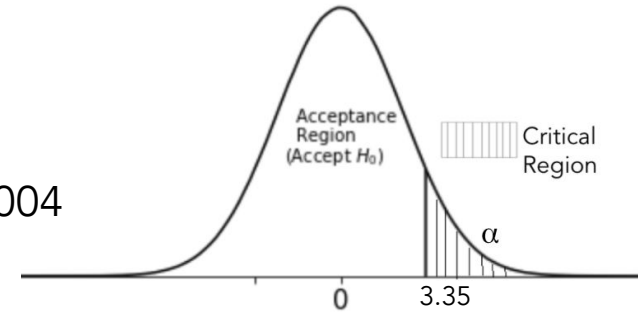
Solution:

The test statistics:
$$Z_{calc} = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} = \frac{75 - 60}{30 / \sqrt{45}} = 3.35$$

The p-value = $P(Z > Z_{calc}, \text{ under } H_0) = P(Z > 3.35, \mu = 60) = 0.0004$

Since p-value < 0.05, we reject H_0 .

Thus there is enough evidence to conclude that the mean delivery time is more than 60 min





Right tailed test

Python solution: Calculate p-value

```
# calculate the corresponding p-value for the test statistic
# use 'sf()' to calculate  $P(Z > 3.35)$ 
p_value = stats.norm.sf(3.35)

print('p-value:', p_value)

p-value: 0.0004040578018640207
```

Since $p\text{-value} < 0.05$, we reject H_0 .

Tests based on Z statistic

Assumptions:

- The data is continuous in nature
- The sample is assumed to be drawn from a population following the normal distribution
- The sample is a simple random sample (each observation is equally likely to be drawn)



Test for normality

- The hypothesis to test whether the population is normally distributed

H_0 : The data is normal

against

H_1 : The data is not normal

- Failing to reject H_0 , implies that the data is normally distributed
- The Shapiro-Wilk test is used

Note: The python code for Shapiro test is `scipy.stats.shapiro(Sample)`

One-Sample test - Z test for Mean (σ known)

A Pizza restaurant claims that the average delivery time of pizza is 20 minutes with the Standard deviation of 12 minutes. A sample of 30 pizza delivery times is given below:

38, 38, 10, 10, 10, 10, 10, 38, 10, 38, 10, 10, 10, 10, 10, 38, 10, 38, 10, 38, 38, 38, 10, 38, 38, 38, 38, 10, 10, 38

Test whether there is a significant deviation between your sample mean and the population mean?

Solution

We observe from the sample, **sample mean** = $692 / 30 = 23.07$

n = 30

μ = 20 and **σ** = 12

One-Sample test - Z test for Mean (σ known)

Step 1: State the null and alternative hypothesis

$$H_o: \mu \leq 20 \text{ minutes}$$

$$H_a: \mu > 20 \text{ minutes}$$

Step 2: Choose the level of significance, α to be 5% and sample size is 30

Step 3: Choose the appropriate test statistic. Since σ is known, you use the normal distribution and ZSTAT as test statistic.

Step 4: Determine the critical value that divides the rejection and non-rejection regions.

Critical values are selected so that the rejection region contains a total area of α when H_o is true and the non-rejection region contains a total area of $1 - \alpha$ when H_o is true. Since $\alpha = 0.05$, the critical value of ZSTAT is **1.65**.

One-Sample test - Z test for Mean (σ known)

Critical value

Step 5: Collect the data, organize the results and compute the value of the test statistic.

$$Z_{STAT} = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

$$Z = (23.07 - 20) / (12 / \sqrt{30}) = 1.40$$

Step 6: State the statistical decision and the managerial conclusion.

- Z_{STAT} is in the region of non-rejection because $Z_{STAT} < Z_{CRITICAL}$; ($Z_{CRITICAL} = 1.65$) where $Z_{STAT} = 1.40$
- Because the test statistic falls in the non-rejection region, we fail to reject the null hypothesis.
- The managerial conclusion is that insufficient evidence exists to prove that pizza delivery time is different from the claimed delivery time of 20 minutes. No Corrective action on delivery process is required.

One-Sample test - Z test for Mean (σ known)

P value approach

Step 5: Collect the data, organize the results and compute the p value of the test statistic.

$$Z = (23.07 - 20) / (12 / \text{sqrt}(30)) = 1.40$$

$$P(X > Z = 1.40) = 1 - 0.9192 = 0.0808$$

Step 6: State the statistical decision and the managerial conclusion.

- Since P-value (0.0808) > 0.05 (our level of significance), we fail to reject null hypothesis.
- The managerial conclusion is that insufficient evidence exists to prove that pizza delivery time is different from the claimed delivery time of 20 minutes. No Corrective action on delivery process is required.



The python code to conduct a Z test for population mean is

```
statsmodels.stats.weightstats.ztest(sample, value, alternative)
```

Summary

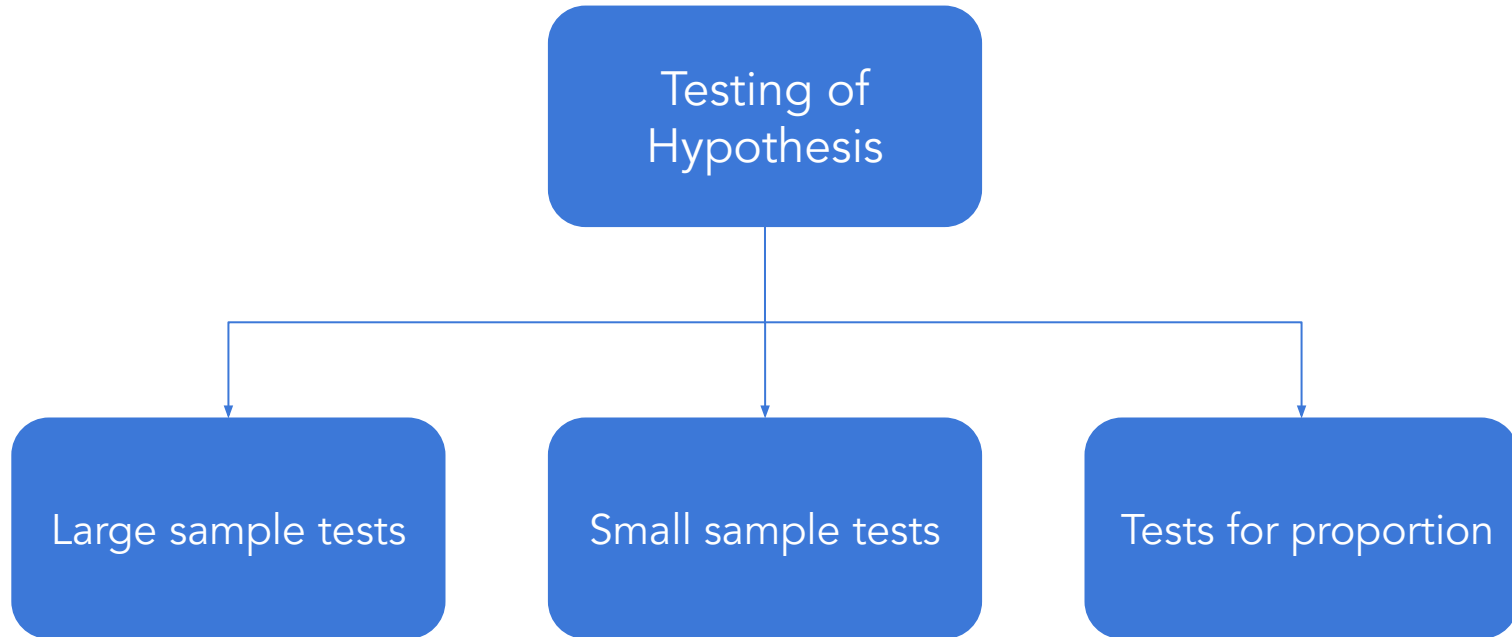
- The Z test can be used if the sample data is drawn from a normally distributed population
- The normality of the dataset can be checked using Shapiro-Wilk test
- Two tailed as well as one tailed (left/ right) Z test can be performed
- The test statistic for the Z test follows normal distribution under the null hypothesis
- The test statistic, p-value and confidence interval can be used to analyze the significance of the test

One Sample – t test

Last topic

- The unknown population parameters are estimated by from sample
- The unknown population parameters are replaced by the sample estimates
- Test the hypothesis for population mean when the population variance is known

In this session...

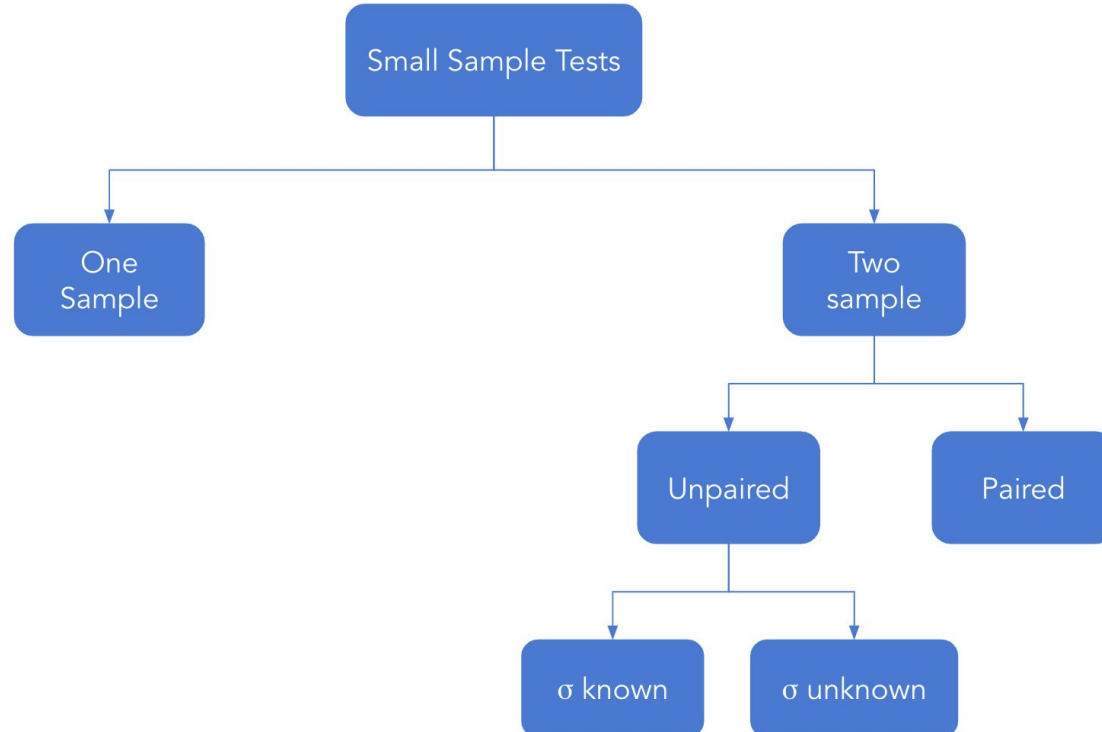


Small Sample Tests

Small sample tests

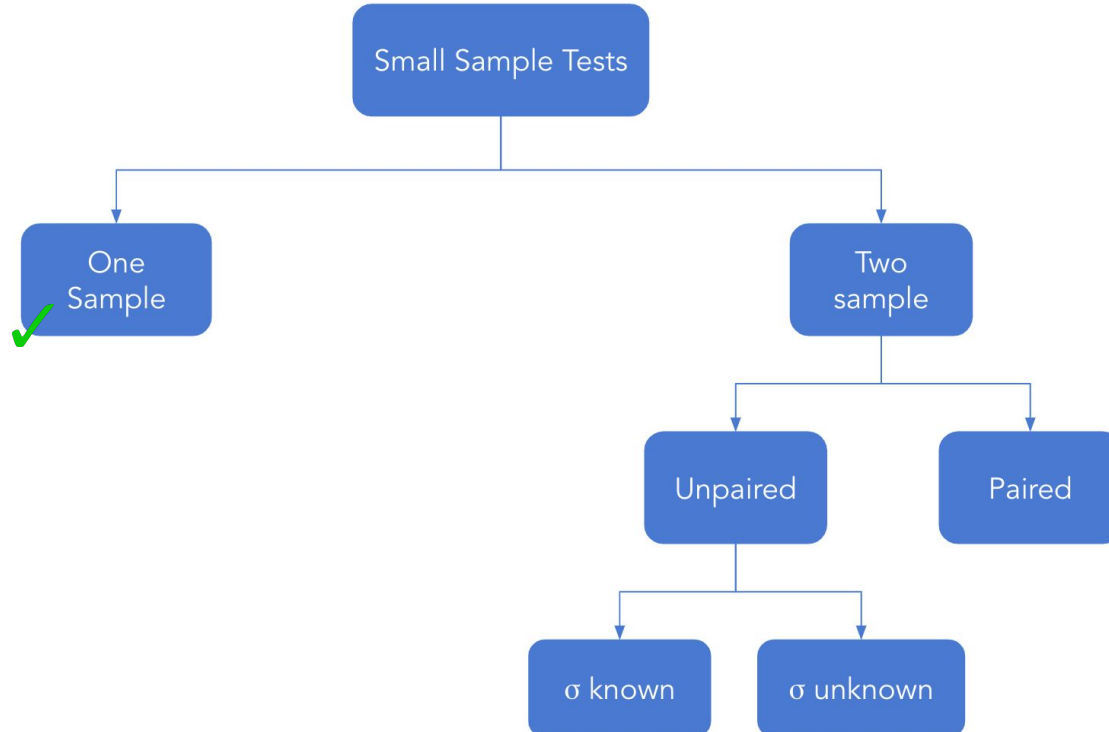
- For tests $n < 30$ are considered to be small sample tests
- Under the null hypothesis the test statistic follow exact distribution - t distribution, F distribution or χ^2 distribution
- Small sample tests are regarded as exact tests

Small sample tests



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Small sample tests



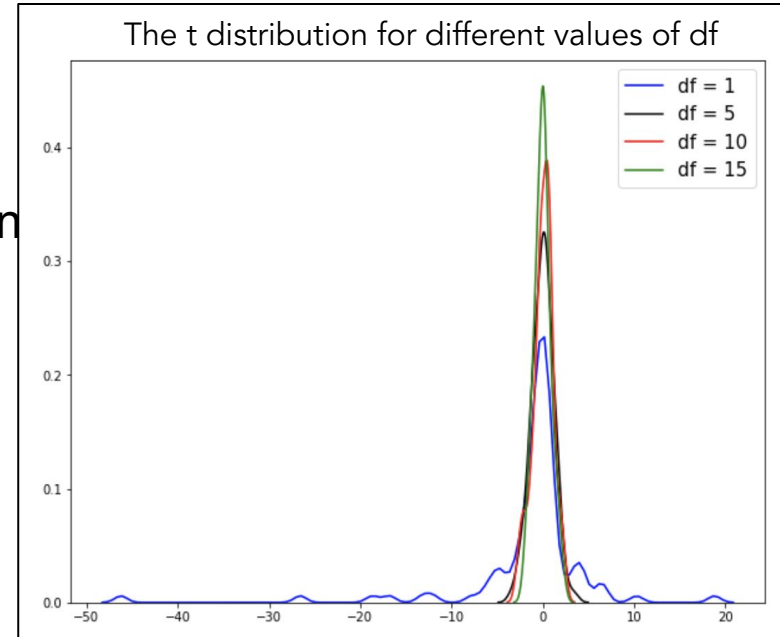
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One sample tests

- To test for population mean, if σ^2 is known the Z test is used
- If σ^2 is unknown, use the t-test
- The t-tests are based on the t-distribution
- The unknown σ is replaced by sample standard deviation (s)

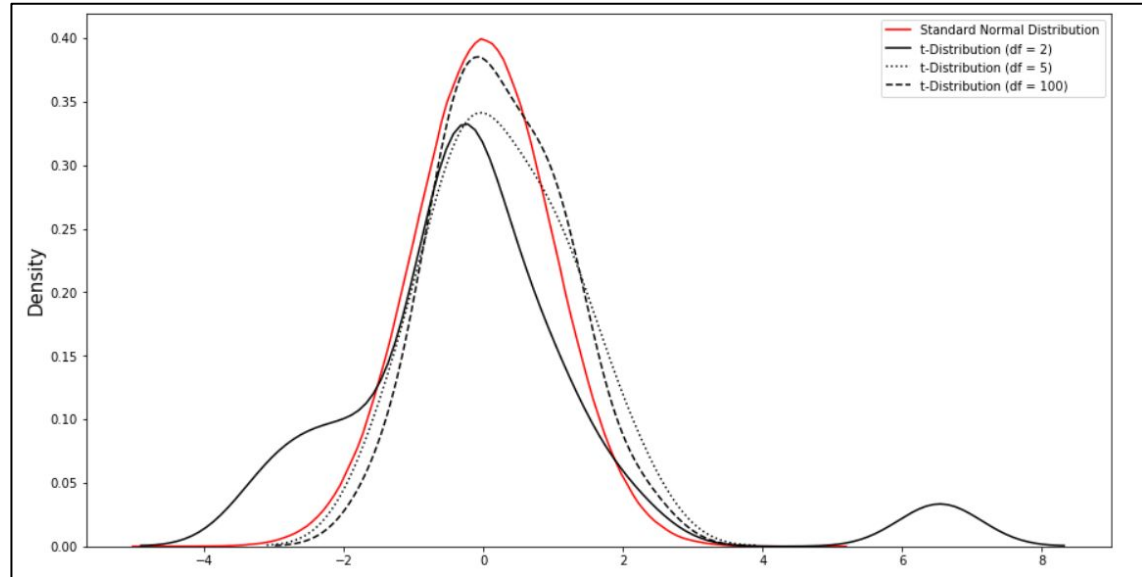
The t-distribution

- It is based on the degrees of freedom
- The t distribution is a symmetric distribution about the mean
- Let n denote the degrees of freedom
- Its mean is 0 and variance is $\frac{n}{n-2}$ (for $n > 2$)



The t-distribution

- It is leptokurtic
- For large n it is mesokurtic



Degrees of freedom

Degrees of freedom (d.f.) is the number of independent variables, i.e. they are free to vary.

Example:

Consider a system of equations:

- $Y = 8 + 5Z$
- $Z = X + 10$

Say if the value of Z is known, it is possible to find the value of other variables. Thus the d.f. is 1.

One sample test - hypothesis

- The hypothesis to test whether the population mean is equal to a specified value when σ is unknown

$$H_0 : \mu = \mu_0 \quad \text{against} \quad H_1 : \mu \neq \mu_0$$

- It implies

H_0 : The population mean is equal to μ_0 (i.e $\mu = \mu_0$)

against H_1 : The population mean is not equal to μ_0 (i.e $\mu \neq \mu_0$)

One sample test - hypothesis

- Like the one sample t- tests, it is possible to test for

$$H_0 : \mu \leq \mu_0 \quad \text{against} \quad H_1 : \mu > \mu_0$$

Or

$$H_0 : \mu \geq \mu_0 \quad \text{against} \quad H_1 : \mu < \mu_0$$

- Failing to reject H_0 implies that the null hypothesis is true

One sample test - test statistic

- The test statistic t given by

$$t = \frac{\bar{X} - \mu}{s / \sqrt{n}}$$

s^2 is the sample estimate of population variance and is given by

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1} \quad \text{or} \quad s^2 = \frac{n(\text{sample variance})}{n-1}$$

- Under H_0 , the test statistic t follows t distribution with $n-1$ degrees of freedom

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The python code to conduct a t test for population mean is

```
scipy.stats.ttest_1samp(Sample, popmean)
```

Decision rule

	H_1	Based on critical region	Based on p-value	Based on confidence interval
For two tailed test	$\mu \neq \mu_0$	Reject H_0 if $ t \geq t_{df, \alpha/2}$	Reject H_0 if p-value is less than or equal to the level of significance	Reject H_0 if μ_0 does not lie in the confidence interval
For left tailed test	$\mu < \mu_0$	Reject H_0 if $t \leq -t_{df, \alpha}$		
For right tailed test	$\mu > \mu_0$	Reject H_0 if $t \geq t_{df, \alpha}$		



One sample t-test

Question:

A researcher is studying the growth of bacteria in waters of Lake Beach. The mean bacteria count of 100 per unit volume of water is within the safety level. The researcher collected 10 water samples of unit volume and found the mean bacteria count to be 94.8 with a sample variance of 72.66. Does the data indicate that the bacteria count is within the safety level? Test at the $\alpha = .05$ level. Assume that the measurements constitute a sample from a normal population.



One sample t-test

Solution:

The mean bacteria count of 100 per unit volume of water is within the safety level.
i.e. $\mu = 100$

Let X: bacteria count per unit volume

The researcher collected 10 water samples of unit volume and found the mean bacteria count to be 94.8 with a sample variance of 72.66.
i.e. $n = 10$, $\bar{X} = 94.8$ and $s^2 = 72.66$



One sample t-test

Solution:

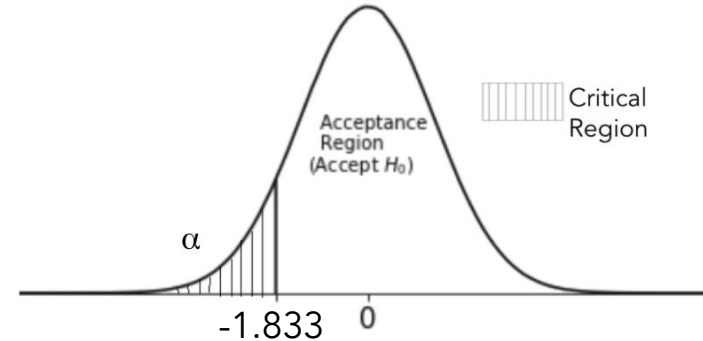
To test: The bacteria count is within the safety level
i.e. $H_0 : \mu \geq 100$ against $H_1 : \mu < 100$

The test statistic is

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}} = \frac{94.8 - 100}{\sqrt{72.66/10}} = -1.929$$

Here t follows $t_{9,0.05}$

From the table we have $t_{9,0.05} = -1.833$



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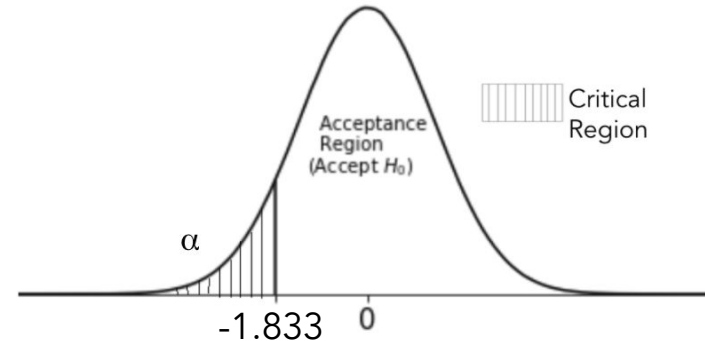
One sample t-test

Solution:

$$t_{\text{critical}} (-1.833) > t_{\text{calc}} (-1.929)$$

Since the critical value is less than the calculated statistic value.

We reject the H_0 , and can conclude that the average bacteria per unit volume (true mean) is within the safety levels.





One sample t-test

Python solution: Calculate critical t-value

```
# calculate the t-value for 95% of confidence level
# use 'stats.t.isf()' to find the t-value corresponding to the upper tail probability 'q'
# pass the value of 'alpha' to the parameter 'q', here alpha = 0.05
# pass the degrees of freedom to the parameter, 'df'
# use 'round()' to round-off the value to 2 digits
t_val = round(stats.t.isf(q = 0.05, df = 9), 2)

print('Critical value for one-tailed t-test:', t_val)

Critical value for one-tailed t-test: 1.83
```

As t-distribution is symmetric, for a left-tailed test if test statistic is less than -1.83, then we reject H_0 .

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One sample t-test

Python solution: Calculate test statistic

```
# define a function to calculate the t-test statistic
# pass the population mean, sample standard deviation, sample size and sample mean as the function input
def t_test(pop_mean, samp_std, n, samp_mean):

    # calculate the test statistic
    t_score = (samp_mean - pop_mean) / (samp_std / np.sqrt(n))

    # return the t-test value
    return t_score

# calculate the test statistic using the function 't_test'
t_score = t_test(pop_mean = 100, samp_std = (72.66)**(0.5), n = 10, samp_mean = 94.8)
print("t-score:", t_score)

t-score: -1.9291040236750068
```

As test statistic ($= -1.929$) < critical value ($= -1.83$), we reject H_0 .

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One sample t-test

Python solution: Calculate p-value

```
# calculate the corresponding p-value for the test statistic
# use 'cdf()' to calculate  $P(t \leq t\_score)$ 
# pass the degrees of freedom to the parameter, 'df'
p_value = stats.t.cdf(t_score, df = 9)

print('p-value:', p_value)

p-value: 0.04289782134327503
```

As $p\text{-value} < 0.05$, we reject H_0 .



Test for population mean (σ unknown)

Question:

The manager of a packaging process at a protein powder manufacturing plant wants to determine if the protein powder packing process is in control. The correct amount of protein powder per box is 350 grams on an average. A sample of 80 boxes was drawn which gave a mean of 354.5 grams with a standard deviation of 15. At 5% level of significance, is there evidence to suggest that the weight is different from 350 grams.



Test for population mean (σ unknown)

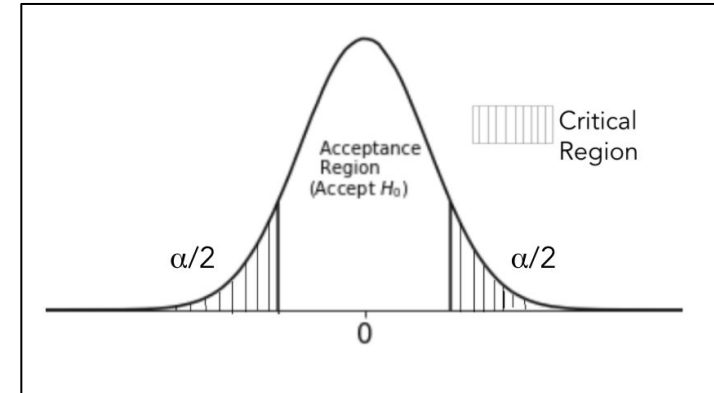
Solution:

X: Amount of protein powder in a box

$$X \sim N(\mu, \sigma^2)$$

Here $n = 80$, $s = 15$, $\mu = 350$, $\bar{X} = 354.5$

To test $H_0: \mu = 350$ against $H_1: \mu \neq 350$





Test for population mean (σ unknown)

Solution:

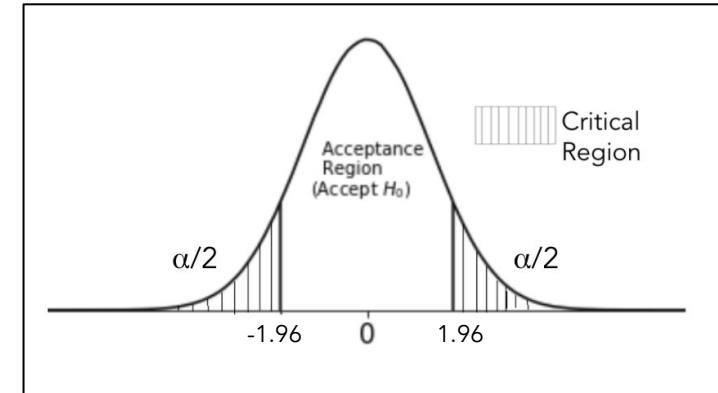
The test statistic is

$$= \frac{\bar{X} - \mu}{s/\sqrt{n}} = \frac{354.5 - 350}{15/\sqrt{80}} = 2.683$$

Decision Rule: Reject $|t_{\text{calc}}| \geq t_{\alpha/2}$

Here $t_{\alpha/2} = 1.96$

Since $2.683 > 1.96$, reject H_0 .



We may conclude that the average weight of protein powder in the box is not 350 grams

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Test for population mean (σ unknown)

Python solution: Calculate critical t-value

i.e. If test statistic is less than -1.96 or greater than 1.96 then we reject H_0 .



Test for population mean (σ unknown)

Calculate test statistic

As test statistic (2.68) > critical value (1.96),
we reject H_0 .



Test for population mean (σ unknown)

solution: Calculate p-value

As p-value < 0.05 ,

we reject H_0 .

One Sample - t test for Mean (σ unknown)

$$t_{STAT} = \frac{\bar{X} - \mu}{s / \sqrt{n}}$$

- where $\bar{X}$ is the mean of the sample
 - μ is the mean of the population
 - s is the Sample standard deviation
 - n is the number of observations
- Here, t follows a t distribution having $n-1$ degrees of freedom

One-Sample test - t test for Mean (σ unknown)

Example: Consider amount from Sales invoice of a nearby restaurant given the population mean is 120 USD.

108.98, 152.22, 111.45, 110.59, 127.46, 107.26, 93.32, 91.97, 111.56, 75.71, 128.58, 135.11

Test whether there is a significant deviation between your sample mean and the population mean?

Solution:

We observe from the sample,

Sample mean = $1354.21 / 12 = 112.85$

Sample Std. deviation = 20.80

$n = 12$

$\mu = 120$

$s = 20.80$

One-Sample test - t test for Mean (σ unknown)

Step 1: State the null and alternative hypothesis

$$H_0: \mu = 120$$

$$H_a: \mu \neq 120$$

Step 2: Choose the level of significance, α to be 5% and sample size is 12

Step 3: Choose the appropriate test statistic. Since σ is unknown, you use the t distribution and t_{STAT} as test statistic.

Step 4: Determine the critical value that divides the rejection and non-rejection regions.

For a given sample size n , the test statistic follows a t distribution with $n-1$ degrees of freedom. The alternative hypothesis has inequality sign and therefore has **two tails**. The area of rejection of the t distribution's left (lower) tail is 0.025, and the area in the rejection region of the t distribution's right (upper) tail is also 0.025. The t critical values are (-2.201 and 2.201) for 11 degrees of freedom at 5 % level of significance.

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One-Sample test - t test for Mean (σ unknown)

Using Critical Value approach

Step 5: Collect the data, organize the results and compute the value of the test statistic.

$$t = (112.85 - 120) / (20.80 / \sqrt{12}) = -1.191$$

Step 6: State the statistical decision and the managerial conclusion.

- t is not in the region of rejection because $t_{\text{STAT}} < t_{\text{CRITICAL}} (-2.201)$ where $t_{\text{STAT}} = -1.191$. Because the test statistic falls in the non-rejection region, the statistical decision is not to reject the null hypothesis.
- The managerial conclusion is that insufficient evidence exists to prove that mean amount per invoice has changed.
- No Corrective action on sales process is required.

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One-Sample test - t test for Mean (σ unknown)

P value approach

Step 5: Collect the data, organize the results and compute the p value of the test statistic.

$$t = (112.85 - 120) / (20.80 / \text{sqrt}(12)) = -1.1908$$

$$P(X > t = -1.1908, 11 \text{ df}) = 0.259$$

Step 6: State the statistical decision and the managerial conclusion.

Since P-value (=0.259) > 0.05 (our level of significance), we do not reject null hypothesis. So statistical decision is not to reject the null hypothesis.

The managerial conclusion is that insufficient evidence exists to prove that mean amount per invoice has not changed.

No Corrective action on sales process is required.

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